Rockets and Feathers in the Egg Aisle

Retail egg prices have followed wholesale prices up faster than down since the bird-flu outbreaks

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Abstract

Retail egg prices have followed wholesale prices up faster than down since the bird-flu outbreaks began in 2015, and over January to October 2026 shoppers paid roughly $2 billion more for eggs than the historical relation between the two prices implies. In an error-correction model of the BLS average price of a dozen eggs on the producer price index for eggs, a wholesale increase passes 0.61 into the retail price within four months and a decrease 0.32, against 0.80 and 0.79 over 1980-2014. The asymmetry survives dropping 2025 and 2026, and it appears in the South, where no state cage-free mandate applies. In October 2026 wholesale prices were 94 percent below their 2025 peak, while the retail price of $2.32 a dozen was 66 percent above its 2019 average, against general inflation of 32 percent. Eggs appear to be the clearest case of a wider change. Since 2015 retail milk, coffee and gasoline prices have also responded more to cost increases than to decreases, while before 2015 only coffee did.

1 The question

Peltzman (2000) found that across many products prices rise faster than they fall, and Borenstein, Cameron and Gilbert (1997) documented the same pattern for gasoline, where it is known as rockets and feathers. Eggs offer an unusually clean test. The wholesale price is volatile, retail eggs are a near-homogeneous product, and since 2015 the market has been hit by three large supply shocks from highly pathogenic avian influenza, in 2015, 2022-23 and 2024-25, each followed by a collapse in wholesale prices.

We use the Bureau of Labor Statistics average retail price of a dozen Grade A large eggs (FRED series APU0000708111) and the producer price index for chicken eggs (WPU0171), monthly from January 1980 to October 2026, with the CPI-U as a deflator in the long-run relation. Single missing months are interpolated, including October 2025, when neither the retail price nor the CPI was published because of the government shutdown. Longer gaps are left missing, and no monthly change is computed across them.

The model is an error-correction model with separate responses to wholesale increases and decreases,

$$\Delta r_t = \alpha + \sum_{k=0}^{3} \left(\beta_k^{+} \, \Delta w_{t-k}^{+} + \beta_k^{-} \, \Delta w_{t-k}^{-} \right) + \sum_{j=1}^{2} \gamma_j \, \Delta r_{t-j} + \theta \left(r_{t-1} - \phi \, w_{t-1} - \delta \, p_{t-1} \right) + \eta \, \Delta p_t + \varepsilon_t,$$

where r, w and p are the logs of the retail price, the wholesale index and the CPI, and Δw^{+} and Δw^{-} are the positive and negative parts of the wholesale change. The long-run coefficients ϕ and δ come from a first-stage regression of r on w and p within each sample. Standard errors are Newey-West with six lags.

2 Results

Table 1 gives the main result. Before 2015, retail prices responded to wholesale increases and decreases alike, within a month and over four months, and equality cannot be rejected at any conventional level. Since 2015, a wholesale increase passes into the retail price about three times as fast as a decrease in the first month, and twice as much over four months, with p-values of 0.03 and 0.002 for equality. After a month, retail prices have already overshot their long-run response to an increase and covered only 62 percent of their long-run response to a decrease.

The asymmetry is not a product of the latest episode. Ending the sample in 2024, before the 2025 peak and the 2026 collapse, gives four-month responses of 0.59 to increases and 0.36 to decreases ($p = 0.004$), and starting in 2010 gives 0.60 and 0.43 ($p = 0.05$). With one lag of wholesale changes the post-2015 totals are 0.40 and 0.16 ($p = 0.001$) and with six lags 0.61 and 0.28 ($p = 0.03$), while before 2015 the same specifications give p-values of 0.43 and 0.65. Over 2000-2014 alone the responses are 0.63 to increases and 0.74 to decreases, if anything the other way round.

	1980-2014	2015-2026
Same-month response to a wholesale increase, β_0^+	0.29 (0.04)	0.23 (0.04)
Same-month response to a wholesale decrease, β_0^-	0.32 (0.03)	0.08 (0.04)
p-value, equal	0.51	0.03
Four-month response to an increase, $\sum_k \beta_k^+$	0.80	0.61
Four-month response to a decrease, $\sum_k \beta_k^-$	0.79	0.32
p-value, equal	0.99	0.002
Share of long-run pass-through after one month, increase	0.78	1.14
Share of long-run pass-through after one month, decrease	0.74	0.62
Long-run elasticity of retail to wholesale, ϕ	0.65	0.41
Error-correction coefficient, θ	−0.083 (0.022)	−0.138 (0.035)
Months	403	138

Table 1: Asymmetric error-correction models of the log retail price of a dozen eggs on the log producer price index for eggs, monthly. Newey-West standard errors (six lags) in parentheses. The shares of long-run pass-through come from simulating the estimated model after a permanent 10 percent wholesale change, as a fraction of the long-run response ϕ; a share above one means the retail price overshoots its long-run response.

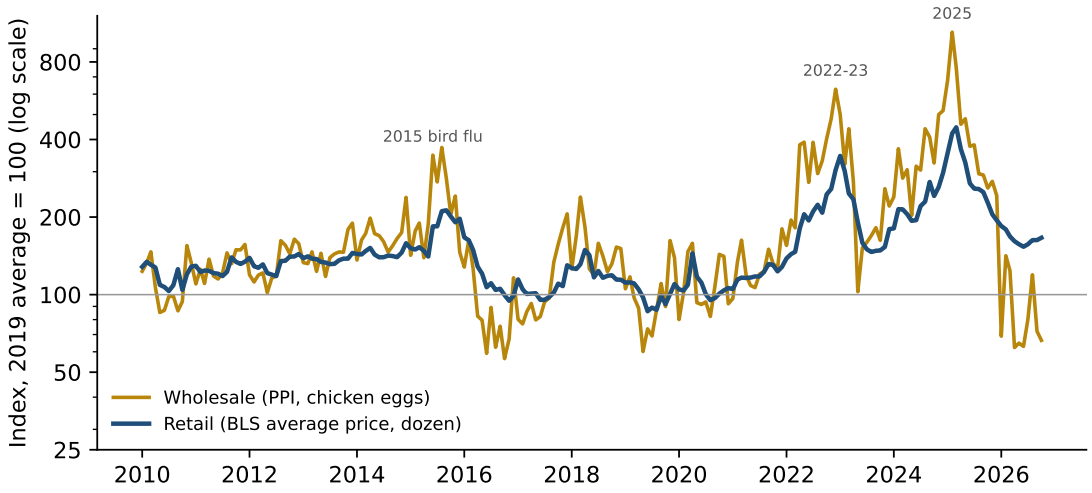


Figure 1: Retail and wholesale egg prices, monthly, each indexed to its 2019 average (log scale). Retail is the BLS average price of a dozen Grade A large eggs; wholesale is the producer price index for chicken eggs.

Figure 1 shows what this means in 2026. After the 2025 outbreak, wholesale egg prices fell 94 percent from their peak, and by October 2026 the index stood about a third below its 2019 average. The retail price fell from $6.23 in March 2025 to $2.32 in October 2026, which is 63 percent below its peak but still 66 percent above its 2019 average of $1.40, while the CPI rose 32 percent over the same period. Relative to the long-run relation estimated over 1980-2026, the retail price in October 2026 is 0.43 log points, or roughly 50 percent, higher than wholesale prices and the general price level imply. The gap reached 0.53 in January 2026, and six of the eight largest monthly gaps since 1980 fall in 2026, the other two in late 2015 and mid-2023.

3 Three checks

Is it eggs? Table 2 applies the same model to other staples whose farm or wholesale costs swing. Since 2015 the response to cost increases exceeds the response to decreases at the 5 percent level for milk, coffee and gasoline as well as eggs, while before 2015 only coffee showed this. Chicken passes the same test only when its single missing month is interpolated, so it is at best a marginal case. Beef shows no asymmetry in either period. For coffee and gasoline the post-2015 response to decreases is negative, so retail prices did not fall at all when costs fell. The pattern is suggestive. The long-run elasticities of retail to cost are small for several of these products, the data are national monthly averages, and in the 2021-2026 subsample

Retail price / cost measure (data start)	Before 2015	2015-2026
Eggs / PPI chicken eggs (1980)	0.80 / 0.79 (0.99)	0.61 / 0.32 (0.002)
Whole milk / PPI raw milk (1995)	0.38 / 0.29 (0.60)	0.29 / 0.13 (0.005)
Ground beef / PPI slaughter cattle (1984)	0.41 / 0.21 (0.19)	−0.08 / 0.21 (0.10)
Whole chicken / PPI slaughter poultry (1980)	0.23 / 0.16 (0.26)	0.08 / −0.10 (0.04)
Ground coffee / arabica price (1992)	0.40 / −0.21 (0.03)	0.16 / −0.35 (0.001)
Gasoline / WTI crude oil (1986)	0.08 / 0.17 (0.41)	0.19 / −0.07 (< 0.001)

Table 2: Four-month responses of the retail price to increases / decreases in the cost measure, from the same error-correction model as Table 1, with the p-value for equality in parentheses. Retail prices are BLS national average prices; cost measures are producer price indices, the IMF arabica price and the WTI spot price, all monthly from FRED. Single missing months are interpolated; without interpolating its one missing month (May 2020), the chicken result for 2015-2026 becomes 0.04 / −0.03 ($p = 0.60$). A negative response to decreases means the retail price did not fall when costs fell.

Region	Before 2015	2015-2024	Gap, latest month
South	0.90 / 0.84 (0.35)	0.68 / 0.49 (0.04)	0.51 (Oct 2026)
Midwest	0.97 / 0.91 (0.48)	0.62 / 0.28 (0.02)	0.23 (Sep 2025)
Northeast	0.46 / 0.69 (0.28)	0.50 / 0.39 (0.35)	0.07 (Oct 2024)
United States	0.80 / 0.79 (0.99)	0.59 / 0.36 (0.004)	0.43 (Oct 2026)

Table 3: Four-month responses of regional BLS average egg prices to increases / decreases in the national egg PPI, with p-values for equality, and the gap in log points between the retail price and the long-run relation in the latest month available. The Northeast series has no data from mid-2006 to April 2016 or from April 2020 to November 2021 and ends in October 2024, and the Midwest series ends in September 2025; the West is not available on FRED. No monthly change is computed across a gap.

only eggs and coffee remain asymmetric at the 5 percent level. The change in eggs therefore looks like the clearest case of a wider shift in how retail prices respond to costs.

Is it the cage-free laws? Mandates requiring cage-free eggs apply in California, Massachusetts, several other western states and Michigan, but in no state of the census South. If the cage-free shift drove the results, the South should look like the pre-2015 market, and Table 3 shows that it does not. Over 2015-2024 southern retail prices passed through 0.68 of wholesale increases and 0.49 of decreases within four months ($p = 0.04$), and in October 2026 the southern retail price of $2.37 stood 0.51 log points above the long-run relation, more than the national gap of 0.43. The Midwest shows the asymmetry as well; the Northeast, with the shortest series, does not.

What it cost. The long-run relation implies a counterfactual retail price for each month. Over January to October 2026 the actual price exceeded it by $0.62 a dozen on average, with a range from $0.25 to $1.06. The Department of Agriculture puts table egg availability at about 22.5 dozen per person in 2026, and a little over half of shell eggs are sold through retail outlets. With 340 million people and a retail share between 50 and 60 percent, retail sales are 320 to 385 million dozen a month, and the excess over the ten months comes to $2.0 to $2.4 billion. The figure inherits the uncertainty of the long-run relation, which is estimated over a period in which the composition of eggs and retail costs changed, and it should be read as an order of magnitude.

4 Interpretation and scope

The long-run elasticity of retail to wholesale prices has also fallen, from 0.65 before 2015 to 0.41 after, so a smaller share of the retail price now moves with the wholesale egg. Part of this is likely the shift toward cage-free eggs required by state laws, such as California's from 2022, which raises costs that the conventional wholesale index does not capture and may change the mix behind the retail average. The southern results show that this shift does not explain the asymmetry or the 2026 gap. Retail purchases on contract, together with packaging, freight and store costs that do not move with the egg, can slow the pass-through of decreases. Why this should have started around 2015, in eggs and in several other products, is not settled here. One candidate is that large, well-publicized cost shocks make price increases easy to justify and decreases easy to delay.

Ajoseh, Thornsbury and Onel (2025) study weekly wholesale and retail egg prices over 2014-2024 and find that retail prices do the adjusting toward the long-run relation while wholesale prices do not, and that transmission changes with the severity of outbreaks. The contribution here is the split between increases and decreases before and after 2015, its appearance in other products and in the South, and the 2026 gap. The data are monthly national averages, the wholesale series is a producer price index and not a transaction price per dozen, and the retail average may mix conventional and cage-free eggs in changing proportions. Nothing here identifies market power or a cause of the asymmetry.

References

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[3] Peltzman, S. (2000). Prices rise faster than they fall. *Journal of Political Economy* 108(3), 466–502.